

TransDreamer

RL With Transformer World Models

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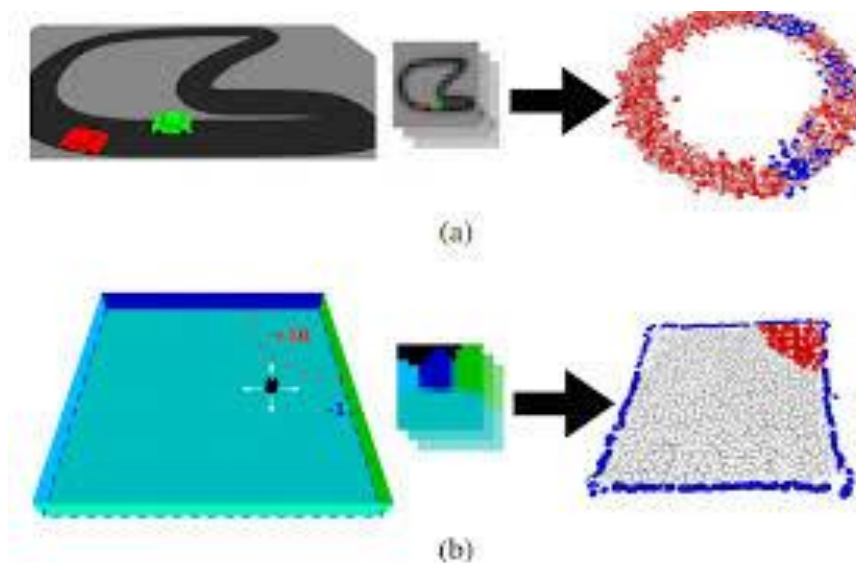
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Introduction

Challenges in Representation Learning in Visual RL

- In visual control problems, unifying the observation representation and task-specific information into single end-to-end training is difficult
 - Conventional model-free methods are confused
 - Because they learn the model and policy using reward solely (TD3, SAC, D4PG, ...)
 - A number of prior works have explored the use of various approaches in RL to learn such representations
 - Learning auxiliary tasks
 - Data augmentation: DrQ
 - Latent dynamics: Flare, DeepMDP
 - Self-supervised learning: Plan2Explore, CURL
 - Previous model-based methods are computationally expensive
 - Because they learn the model and policy separately (PlaNet, SimPle, ...)
 - However, RNN-based model-based methods showed better performance than others

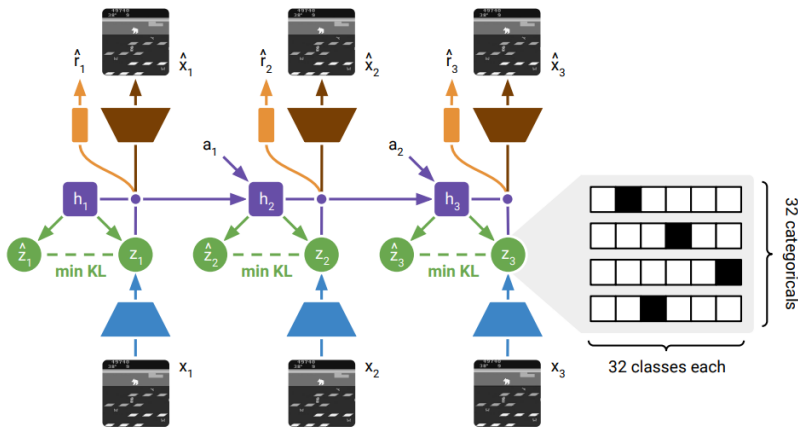


Representation learning in RL

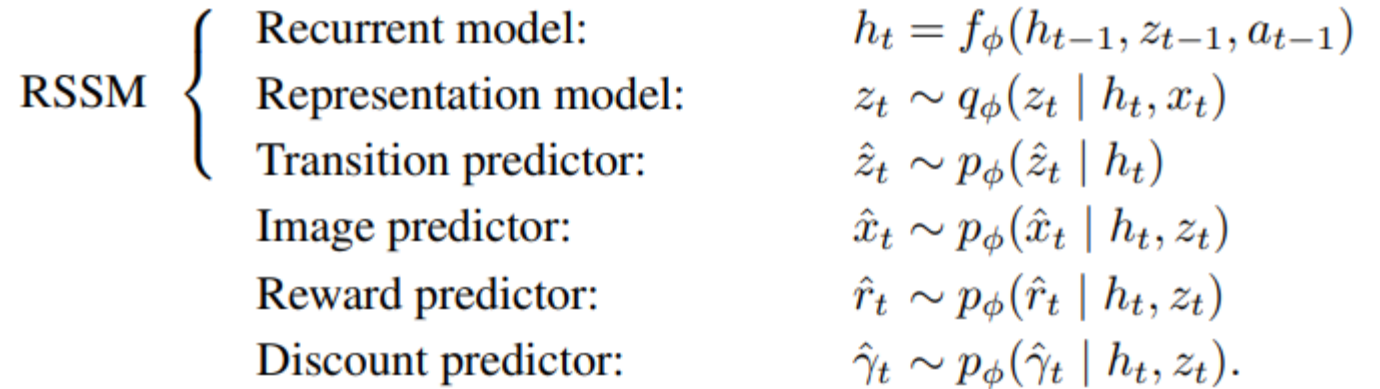
TransDreamer

DreamerV2 (I) – World model

- The first Model-Based Reinforcement Learning(MBRL) agent that achieves human-level performance on the Atari Learning Environment(ALE)
 - The world model consists of an **image encoder**, a **Recurrent State-Space Model(RSSM)** to learn the dynamics, and **predictors** for the image, reward, and discount factor
 - RSSM represents a latent state s_t by the concatenation of a stochastic state z_t and a deterministic state h_t which are updated by $z_t \sim p(z_t|h_t)$ and $h_t = f_{\text{RNN}}(h_{t-1}, z_{t-1}, a_{t-1})$, respectively
 - Deterministic path helps to model the temporal dependency in the world model and stochastic state makes it possible to capture the stochastic nature of the world
 - Using the above models, rollouts can be executed efficiently in a compact latent space without the need to generate observation images.



World model learning sequence



The model components

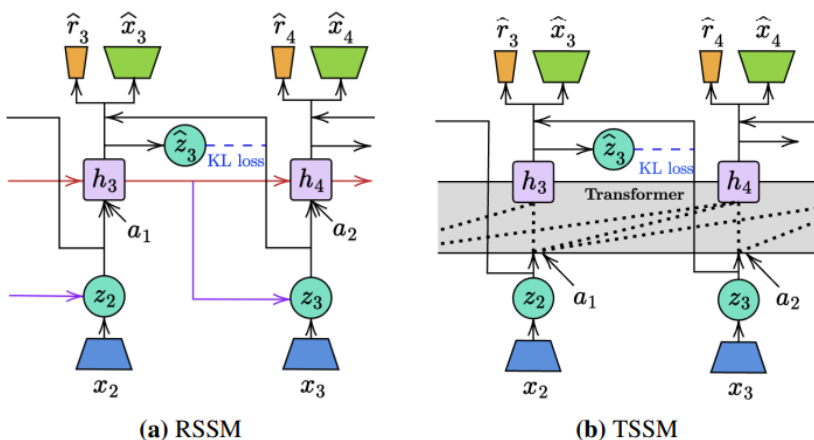
● Overview of the TransDreamer

- Unlike the model-based RL methods that learn world model or dynamics through existing MLP-based or RNN-based models that have inherent limitations, TransDreamer uses a Transformer-based model that has recently shown good performance in diverse tasks to solve more complex tasks
- They specifically substituted the RNN-based backbone to the Transformer-based backbone in the Dreamer framework for solving long-term memory-based reasoning
 - They demonstrated a superior ability to capture long-term dependency than RNN-based Dreamer through experiment
- They proposed a transformer-based action-conditioned model for predicting the observation, reward, and discount, and optimized the objective function as follows
 - $p(o_t, z_{1:T} | a_{1:T}) = \prod_t p(o_t | h_t, z_t) p(z_t | z_{1:t-1}, a_{1:t-1})$, where $o_t = (x_t, r_t, \gamma_t)$ and $h_t = f_{\text{transformer}}(z_{1:t-1}, a_{1:t-1})$
 - $\text{ELBO} = \sum_{t=1}^T (\mathbb{E}_{\prod_{\tau=1}^{t-1} q(z_\tau | x_\tau)} [\ln p(x_t | h_t, z_t) + \ln p(r_t | h_t, z_t) + \ln p(\gamma_t | h_t, z_t) - D_{\text{KL}}[q(z_t | x_t) || p(z_t | z_{1:t-1}, a_{1:t-1})]])$
- Evaluation demonstrated that TransDremaer has outperformed both long-term memory tasks and short-term memory tasks in terms of final performance and qualitative imagination than Dreamer

TransDreamer (II) – RSSM to TSSM

● Transformer-based MBRL agent inherits from the Dreamer framework

- The authors introduced the Transformer State Space Model(TSSM) as the first transformer-based stochastic world model
 - Beyond the simple replacement of RNN to Transformer, the following effects are obtained
 - In any step, the TSSM could directly access past states
 - The TSSM could update the states of each step in parallel during training
 - Furthermore, it retains the following advantageous characteristics of RSSM
 - TSSM could still roll out sequentially for trajectory imagination at test time
 - Proposed TSSM is still a stochastic latent variable model



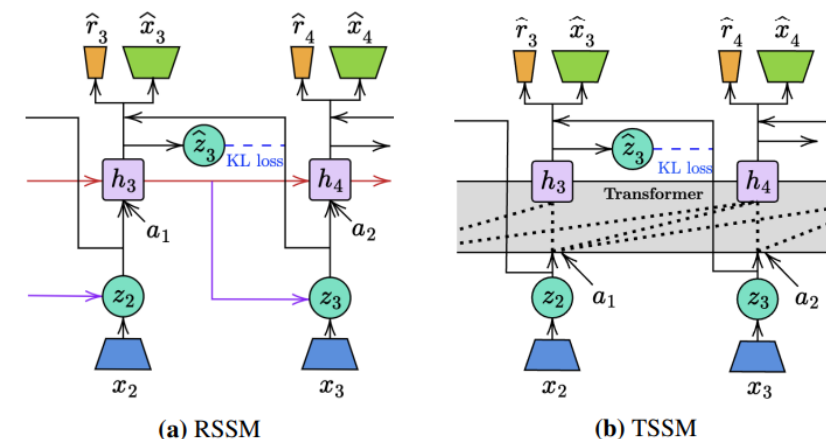
Architecture of the RSSM and TSSM

	RSSM	TSSM
Deterministic state model	$h_t = \text{gru}(h_{t-1}, z_{t-1}, a_{t-1})$	$h_t = \text{transformer}(z_{1:t-1}, a_{1:t-1})$
Representation model	$z_t \sim q(z_t h_t, x_t)$	$z_t \sim q(z_t x_t)$
Stochastic state model	$\hat{z}_t \sim p(\hat{z}_t h_t)$	
Image predictor	$\hat{x}_t \sim p(\hat{x}_t h_t, z_t)$	
Reward predictor	$\hat{r}_t \sim p(\hat{r}_t h_t, z_t)$	
Discount predictor	$\hat{\gamma}_t \sim p(\hat{\gamma}_t h_t, z_t)$	

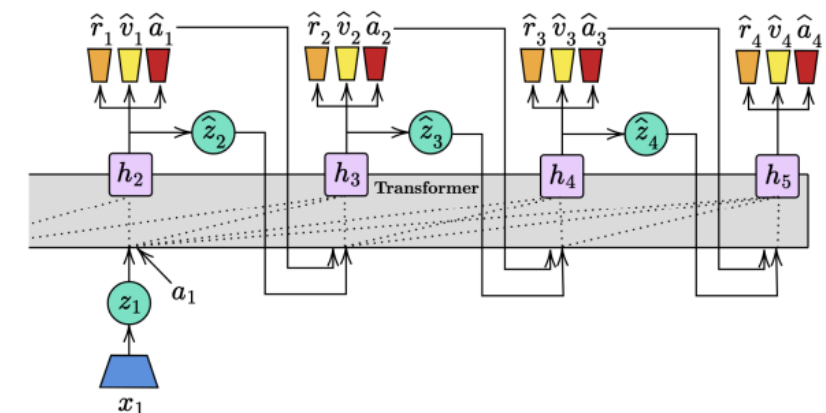
Comparison of the Component Models of RSSM and TSSM

● About TSSM

- Myopic representation model
 - They proposed approximating the posterior representation model by $q(z_t|x_t)$, removing h_t
- Imagination
 - During imagination, they used the prior stochastic state $\hat{z}_t \sim p(\hat{z}|h_t)$ as the input to the transformer to autoregressively generate future states
- Policy learning
 - The policy learning in TransDreamer inherited the general framework of Dreamer
- Number of Imagination trajectories
 - Due to the increased memory requirements of transformers compared with RNNs, they randomly choose a smaller subset of starting states of size K to generated imagined trajectories from



Architecture of the RSSM and TSSM



Policy learning in TSSM

● TSSM Objective function

- The authors optimized the following objective, which is the negative ELBO of the action-conditioned model with additional terms for predicting the reward and discount,

- $$\mathcal{L}_{\text{TSSM}}(\phi) = \sum_{t=1}^T \left(\mathbb{E}_{\Pi_{\tau=1}^t q_{\phi}(z_{\tau}|x_{\tau})} [-\eta_x \ln p_{\phi}(x_t|h_t, z_t) - \eta_r \ln p_{\phi}(r_t|h_t, z_t) - \eta_{\gamma} \ln p_{\phi}(\gamma_t|h_t, z_t)] + \mathbb{E}_{\Pi_{\tau=1}^{t-1} q_{\phi}(z_{\tau}|x_{\tau})} [D_{\text{KL}}[q_{\phi}(z_t|x_t) || p_{\phi}(z_t|z_{1:t-1}, a_{1:t-1})]] \right), \quad \eta_* \text{ are hyper params}$$

- The action-conditioned generative model is

- $p(o_t, z_{1:T} | a_{1:T}) = \Pi_t p(o_t|h_t, z_t) p(z_t | z_{1:t-1}, a_{1:t-1}), \quad \text{where } o_t = (x_t, r_t, \gamma_t) \text{ and } h_t = f_{\text{transformer}}(z_{1:t-1}, a_{1:t-1})$

- By approximating the posterior by $q(z_t|x_t)$, a variational posterior is $q(z_{1:T}|o_{1:T}, a_{1:t-1}) = \Pi_t q(z_t|x_t)$. Thus,

- $$\begin{aligned} \ln p(o_{1:T}|a_{1:T}) &= \ln \mathbb{E}_{p(z_{1:T}|o_{1:T}, a_{1:T})} [\Pi_{t=1}^T p(o_t|h_t, z_t)] = \ln \mathbb{E}_{q(z_{1:T}|o_{1:T}, a_{1:T})} [\Pi_{t=1}^T p(o_t|h_t, z_t) p(z_t|z_{1:t-1}, a_{1:t-1}) / q(z_t|x_t)] \\ &\geq \mathbb{E}_{\Pi_{t=1}^T q(z_t|x_t)} [\sum_{t=1}^T \ln p(o_t|h_t, z_t) + \ln p(z_t | z_{1:t-1}, a_{1:t-1}) - \ln q(z_t|x_t)] \\ &= \sum_{t=1}^T \left(\mathbb{E}_{\Pi_{\tau=1}^{t-1} q(z_{\tau}|x_{\tau})} [\ln p(o_t|h_t, z_t)] - \mathbb{E}_{\Pi_{\tau=1}^{t-1} q(z_{\tau}|x_{\tau})} [D_{\text{KL}}[q(z_t|x_t) || p(z_t|z_{1:t-1}, a_{1:t-1})]] \right) \\ &= \sum_{t=1}^T \left(\mathbb{E}_{\Pi_{\tau=1}^{t-1} q(z_{\tau}|x_{\tau})} [\ln p(x_t|h_t, z_t) + \ln p(r_t|h_t, z_t) + \ln p(\gamma_t|h_t, z_t)] - \mathbb{E}_{\Pi_{\tau=1}^{t-1} q(z_{\tau}|x_{\tau})} [D_{\text{KL}}[q(z_t|x_t) || p(z_t|z_{1:t-1}, a_{1:t-1})]] \right), \\ &\text{where } p(o_t|h_t, z_t) = p(x_t|h_t, z_t) p(r_t|h_t, z_t) p(\gamma_t|h_t, z_t) \end{aligned}$$

Experiments

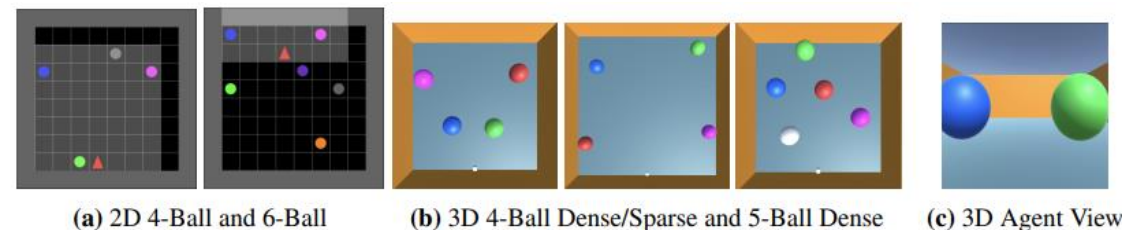
Experiment environment

- The authors tried to answer the following three questions

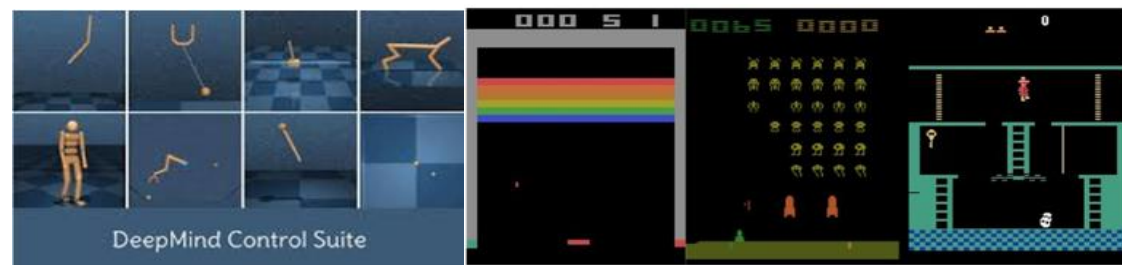
- 1) How do TransDreamer and Dreamer perform in tasks that require long-term memory and reasoning?
 - They made new set of tasks: Hidden order discovery environment
- 2) How do the learned world models of TransDreamer and Dreamer compare?
 - They thoroughly analyze the quality of the world model both quantitatively and qualitatively
- 3) Can TransDreamer also work comparably to Dreamer in environments that require short-term memory?
 - They compared TransDreamer and Dreamer on some tasks in DeepMind Control Suite(DMC) and Atari Learning Env(ALE)

- Environment

- Hidden Order Discovery environment
 - The hidden order of catching balls is fixed in the episode(100 step)
 - Correct order: +3 reward at first visit
- DeepMind Control Suite
 - Environment for visual continuous control(four tasks)
 - Cheetah run, Cup catch, pendulum swing up, walker walk
- ALE
 - Environment for discrete control(four tasks)
 - Boxing, Freeway, Pong, and Tennis tasks



Hidden Order Discovery Environments

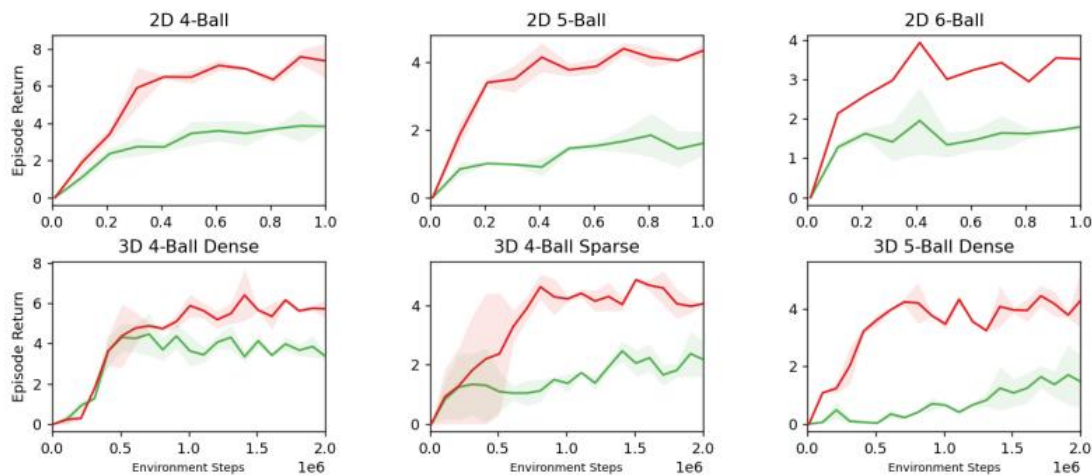


DeepMind Control Suit and Atari Learning Env

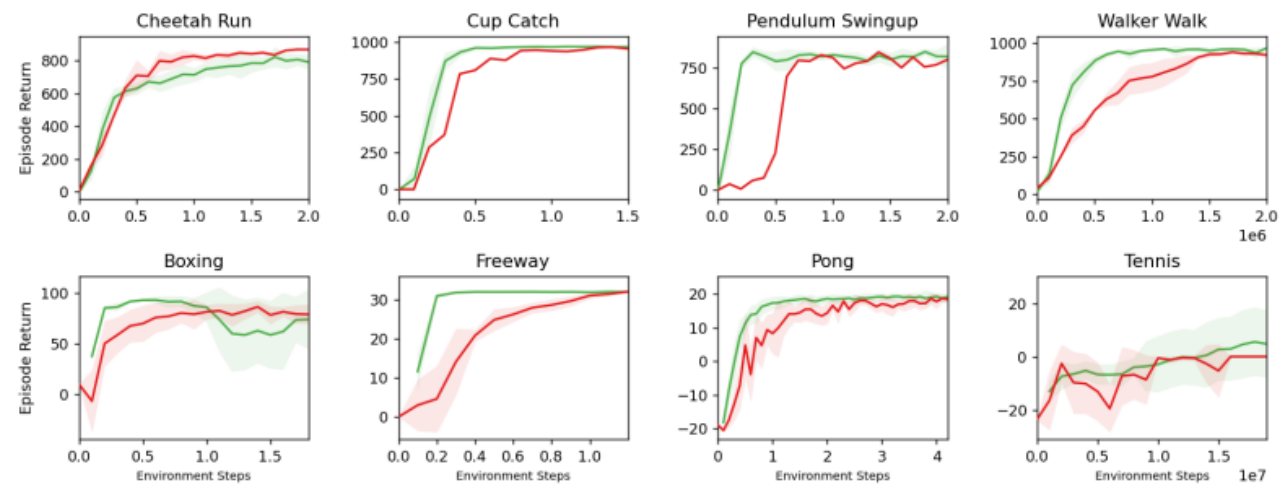
Experiment results (I)

● Comparison with Dreamer

- Experiments show better final performance compared to Dreamer-V2 in long-term memory test (Hidden Order Discovery) and short-term memory test (DMC and ALE).
 - TransDreamer**  **Dreamer** 
- The authors also measure the success rate of each agent in 2-D hidden order discovery tasks
 - For the 4-ball configuration, **TransDreamer** had a success ratio of **23%** and **Dreamer** had only **7%** of that
 - For the 5-ball and 6-ball configuration, **TransDreamer** had success ratios of **5%** and **1%**, and **Dreamer** had **0%** of that



Episode return in Hidden Order Discovery tasks



Episode return in DMC and ALE

Experiment results (II)

● Quantitative and qualitative comparisons with Dreamer

- Then they compare the quality of the trajectories imagined by the TSSM and the RSSM by measuring the generation performance quantitatively and qualitatively.

■ Quantitative results

- They reported the MSE of the predicted images and the reward prediction accuracy during the generation
 - **MSE of predicted image:** TransDreamer generally achieved lower or comparable MSE
 - **Reward prediction:** TransDreamer generally obtained more accurate reward prediction than Dreamer

■ Qualitative results

- They demonstrated the imagined trajectories from TransDreamer and Dreamer in the 5-Ball Dense env
 - The agent and ball reset to their original position in 48 for TransDreamer and 59 for Dreamer

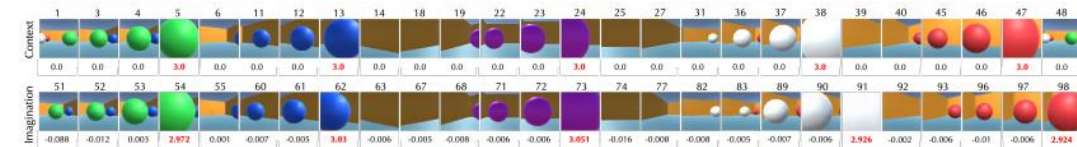
(a) Image Generation Foreground MSE

Task	Model	Context		
		60	70	80
4-Ball Dense	TransDreamer	211.2	133.1	69.8
	DreamerV2	281.9	194.2	110.8
4-Ball Sparse	TransDreamer	195.5	115.2	56.8
	DreamerV2	215.8	138.6	72.4
5-Ball Dense	TransDreamer	245.2	163.1	85.0
	DreamerV2	300.9	217.0	124.9

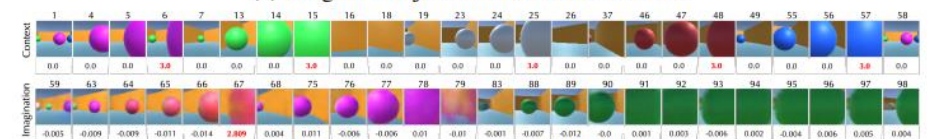
(b) Reward Prediction Accuracy

Task	Model	Context		
		60	70	80
4-Ball Dense	Transdreamer	46.9	53.2	73.2
	DreamerV2	28.2	34.6	50.5
4-Ball Sparse	Transdreamer	32.4	36.5	48.6
	DreamerV2	32.0	33.3	42.3
5-Ball Dense	Transdreamer	17.7	18.1	32.3
	DreamerV2	9.8	6.2	15.3

World Model quantitative comparison



(a) Imagined Trajectories from TransDreamer



(b) Imagined Trajectories from Dreamer

Imagined trajectories comparison

Thank you!

Q&A

- **The first Model-Based Reinforcement Learning(MBRL) agent that achieves human-level performance on the Atari Learning Environment(ALE)**
 - Policy learning is done without interaction with the actual environment; it uses imagined trajectories obtained by simulating the learned world model.
 - Specifically, from each state s_t obtained from a batch sampled from the replay buffer, it generates a future trajectory of length H using the RSSM world model and the current policy as the behavior policy for the imagination.
 - Then, for each state in the trajectory, the rewards $p_\theta(r_t|s_t)$ and the values $v_\psi(s_t)$ are estimated. This allows us to compute the value estimate $V(s_t)$, e.g., by the discounted sum of the predicted rewards and the bootstrapped value $v(s_{t+H})$ at the end of the trajectory.
 - Learning the policy in Dreamer means updating two models, the policy $\pi_\phi(a_t|s_t)$ and the value model $v_\psi(s_t)$.
 - For updating the policy, Dreamer uses the sum of the value estimates of the simulated trajectories, $\sum_{\tau=t}^{t+H} V(s_\tau)$, to construct the objective function.